

Comparing Mitigation Methodologies for Emergent Novel Biases in LLM Decision-Making

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Context

Wu et al. (2025) provide an account of the exploration-exploitation dilemma centered on stereotype formation in sequential decision-making environments. In this sense, social beliefs may not emerge exclusively from pre-existing associations, but from constrained sampling: when early observations presume to favor a particular group-role pairing, decision-makers may exploit that apparent regularity, thus reducing exploration of alternative options (see Bai, Fiske & Griffiths, 2022). Over time, these initial observations can evolve to structured impressions about entire groups. Recent work extends this framework to large language models (LLMs), showing their capacity to develop novel social biases about artificial demographic groups: “LLMs demonstrate a new kind of bias —the creation of novel stereotypes— which manifests over repeated interactions in stateful frameworks” (Bai et al., 2025, p. 10). Most surprisingly, they report that LLMs appear to be “even more prone than humans to develop such biases, even when underlying differences do not exist” (Bai et al., 2025, p. 10). These biases manifest as stratified task allocations, which are more prominent in newer frontier models despite their promise for impartiality; effect presumably caused by higher optimizing power towards given objectives. This suggests that LLMs may not just reproduce inherited social biases, but also generate new ones through adaptive interaction.

Summary of the Proposal

The proposed research investigates the effectiveness of two different strategies for reducing novel emergent biases in LLM sequential allocation tasks, compared to the baseline measurements shown in Wu et al. (2026).

The first strategy is adversarial debiasing, which is a technique that employs a predictive (adversary) model to recover sensitive attributes from a model’s outputs. If the adversary cannot successfully identify the sensitive attribute, then it could be assumed that group-related information is less available to the model’s decision process. The proposed framework introduces an adversary into a sequential allocation task in which an LLM assigns equally qualified candidates from artificial groups to job categories across repeated rounds. Periodically, the adversary observes the LLM’s decisions and explanations with explicit group labels removed and attempts to infer the candidate’s group identity. If group identity becomes inferable from these outputs, this would suggest that the LLM has begun to encode differentiated treatment toward groups despite the absence of real group differences. The adversary’s predictions can then serve both as a diagnostic signal for emerging stratification and as a corrective mechanism: when group identity becomes increasingly recoverable, the intervention redirects the model toward insufficiently explored group-role pairings before early allocation patterns become stable biases.

The second strategy is reward modeling. When a reward function is difficult to specify directly, a model can learn an alternative reward from human preference comparisons (similar to RLHF) and optimize behavior from learned signals. In this setting, reward modeling will be adapted to the sequential allocation task; after the LLM performs the assignments, a reward model will evaluate whether the decision balances performance, fairness and exploration. The question is whether this reward signal will indeed prevent the formation of novel group associations, or if instead, by repeatedly evaluating the model's reasoning alongside group labels, it will enhance implicit associations.

Thus, the study will compare exploration-based prompting as the baseline intervention against two alternative approaches: adversarial debiasing and reward-model-based debiasing. The project will first replicate the baseline intervention using closed models API calls; then, the two alternatives will be tested under the same task structure, job categories, evaluation metrics, etc. The first stage will use closed models such as GPT, Claude or Gemini because of their easy API call access. Subsequently, an open weight model exploration will be performed with models such as Llama or Kimi to examine whether the observed patterns in the closed models can be generalized. Finally, a deeper analysis of the mechanisms behind mitigation will be performed during the final weeks of the project.

Relevance

In terms of relevance, this study connects recent work on dynamic bias formation in LLMs with ongoing debates about how bias should be mitigated in sequential decision-making systems. While prior work has been focused on detecting or reducing biases already present in training data (see also Yang et al. & Zhang et al.), the exploration based account discussed above suggests that bias can also emerge through interaction, even when no real group differences exist. In LLM decision-making, this means that repeated choices and feedback may eventually produce group-role associations that were not originally present.

The proposed study extends this work by comparing different strategies for mitigating such emergent biases. Rather than assuming that any of the three methods is the best solution, the project provides an important comparison to approach bias mitigation in line with Wu et al.'s paradigms. Reward modeling contributes an alignment-based approach for testing whether learned signals reduce bias or make group-role associations more prominent. Adversarial debiasing, despite being effortful and mostly effective only when these biases are very hard to see in the eyes of humans, contributes a corrective signal by detecting when outputs begin to encode group identity; then, redirection and further exploration can be ensured before associations stabilize.

As we enter the so-called agentic era, LLMs become autonomous and become useful in systems that remember past interactions, receive feedback, and make repeated recommendations and allocations. In this context, fairness may arise from the way models form beliefs over time. Therefore, by comparing adversarial debiasing, reward modeling and simple pro-exploration prompting, we can evaluate which interventions can best redirect these dynamics before early patterns become increasingly stable stereotypes.

References

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